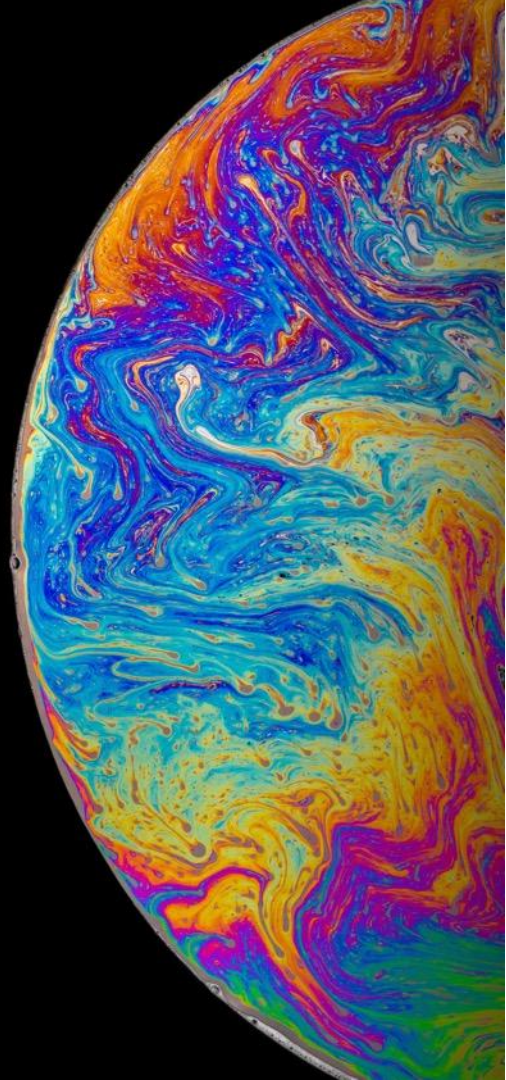
CMIP
Coupled Model Intercomparison Project

ESMValTool
Earth System Model Evaluation Tool

Climate data analysis with the Earth System Model Evaluation Tool

ESMValTool

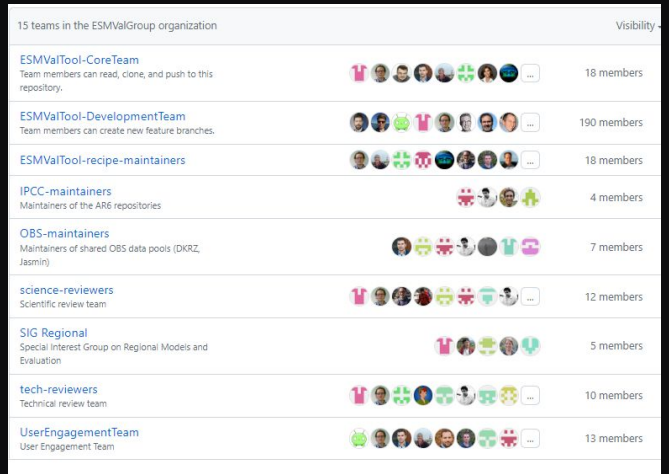


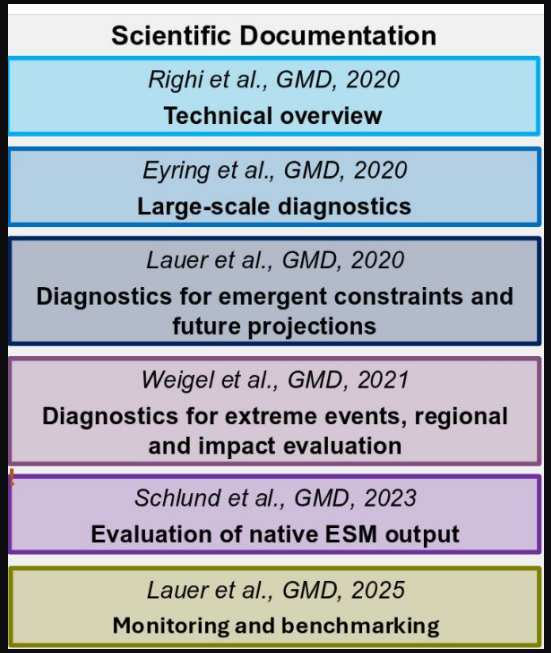
11 November 2026 - Online Seminar - Fresh Eyes on CMIP

What is ESMValTool?

Earth System Model Evaluation Tool (ESMValTool) is a community diagnostic and performance metrics tool for the evaluation of Earth system models (ESMs) participating in the Coupled Model Intercomparison Project (CMIP).

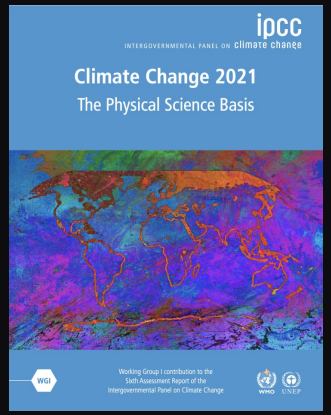
International community effort with ~63 participating institutions and ~200 developers





What is ESMValTool?

Earth System Model Evaluation Tool (ESMValTool) is a community diagnostic and performance metrics tool for the evaluation of Earth system models (ESMs) participating in the Coupled Model Intercomparison Project (CMIP).



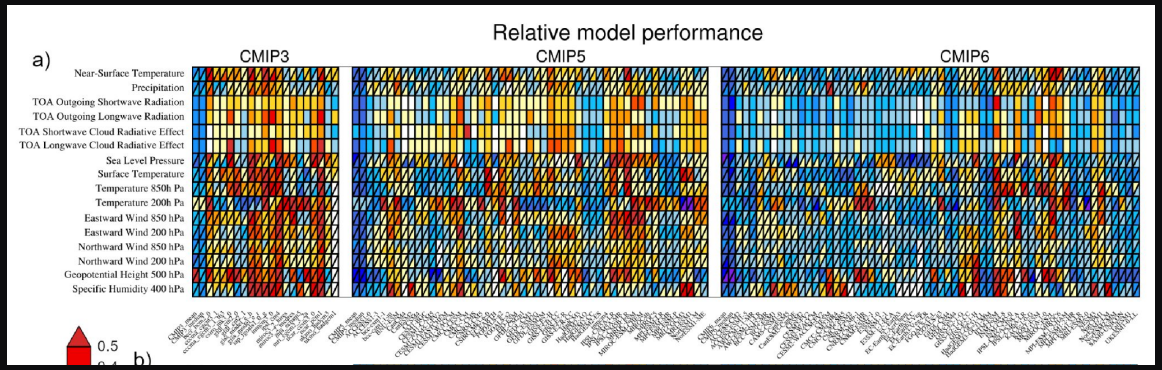


Figure 3.42 in IPCC, 2021: Chapter 3. In: *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change.*

Why to consider ESMValTool for your analysis?

- Includes **many diagnostics** covering different aspects of the Earth system
- **Reproducibility** and of the results
- Supports **several programming languages** (Python, NCL, R, Julia)
- Helps **efficient** data processing
- **Well-documented** source code and diagnostics

Extensive documentation and community support

Tutorial
https://tutorial.esmvaltool.org

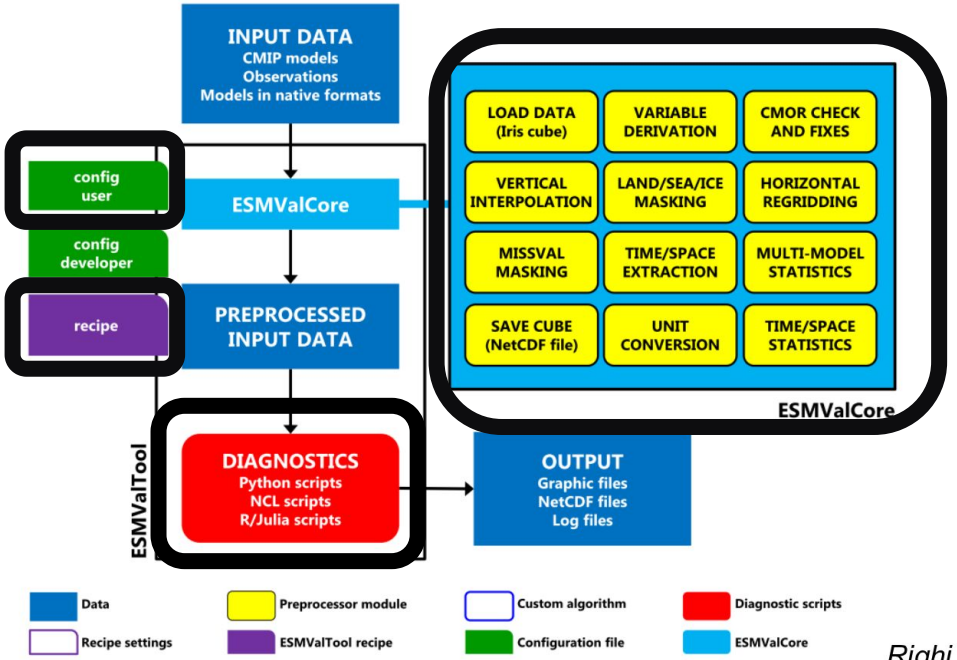
GitHub
https://github.com/ESMValGroup

Website
https://www.esmvaltool.org

Documentation
https://docs.esmvaltool.org

Youtube
https://www.youtube.com/watch?v=sidM4EB6Sbo

Core understanding of ESMValTool



Configuration

Preprocessors

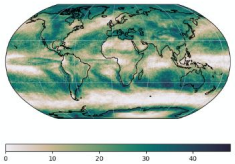
Recipe

Diagnostic

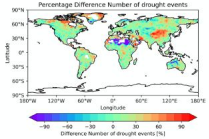
Righi et al., GMD (2020)

CMIP

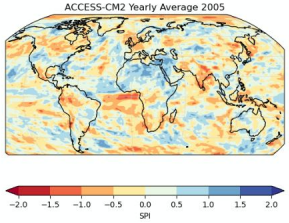
That’s what you can do/reproduce with ESMValTool



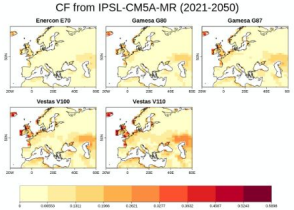
Example of the number of occurrences with consecutive dry days of more than five days in the period ...



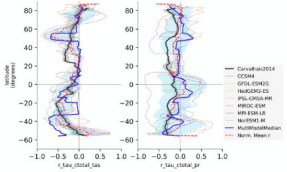
Global map of the percentage difference between multi-model mean of 15 CMIP models and the CRU data ...



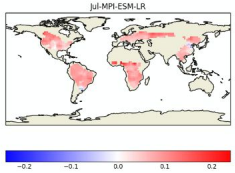
Example plot of SPI averaged over the year 2005. The reference period for index calibration is 2000-...



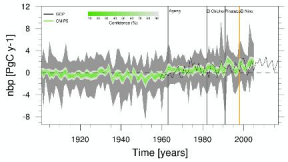
Wind capacity factor for five turbines: Enercon E70 (top-left), Gamesa G80 (middle-top), Gamesa G87 ...



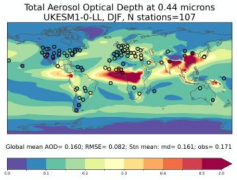
Comparison of latitudinal (zonal) variations of pearson correlation between turnover time and climat...



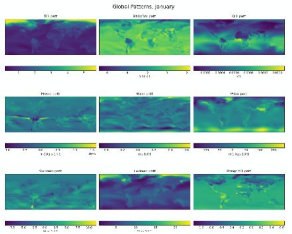
Example of albedo change from tree to crop and grass for the CMIP5 model MPI-ESM-LR derived for the ...



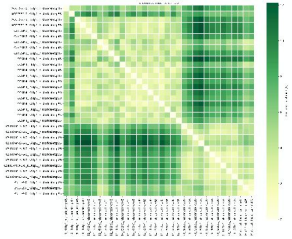
Time series of global net biome productivity (NBP) over the period 1901-2005. Similar to Anav et al....



Evaluation of AOD at 440 nm from UKESM1 historical ensemble member r1i1p1f2 against the AeroNET clim...

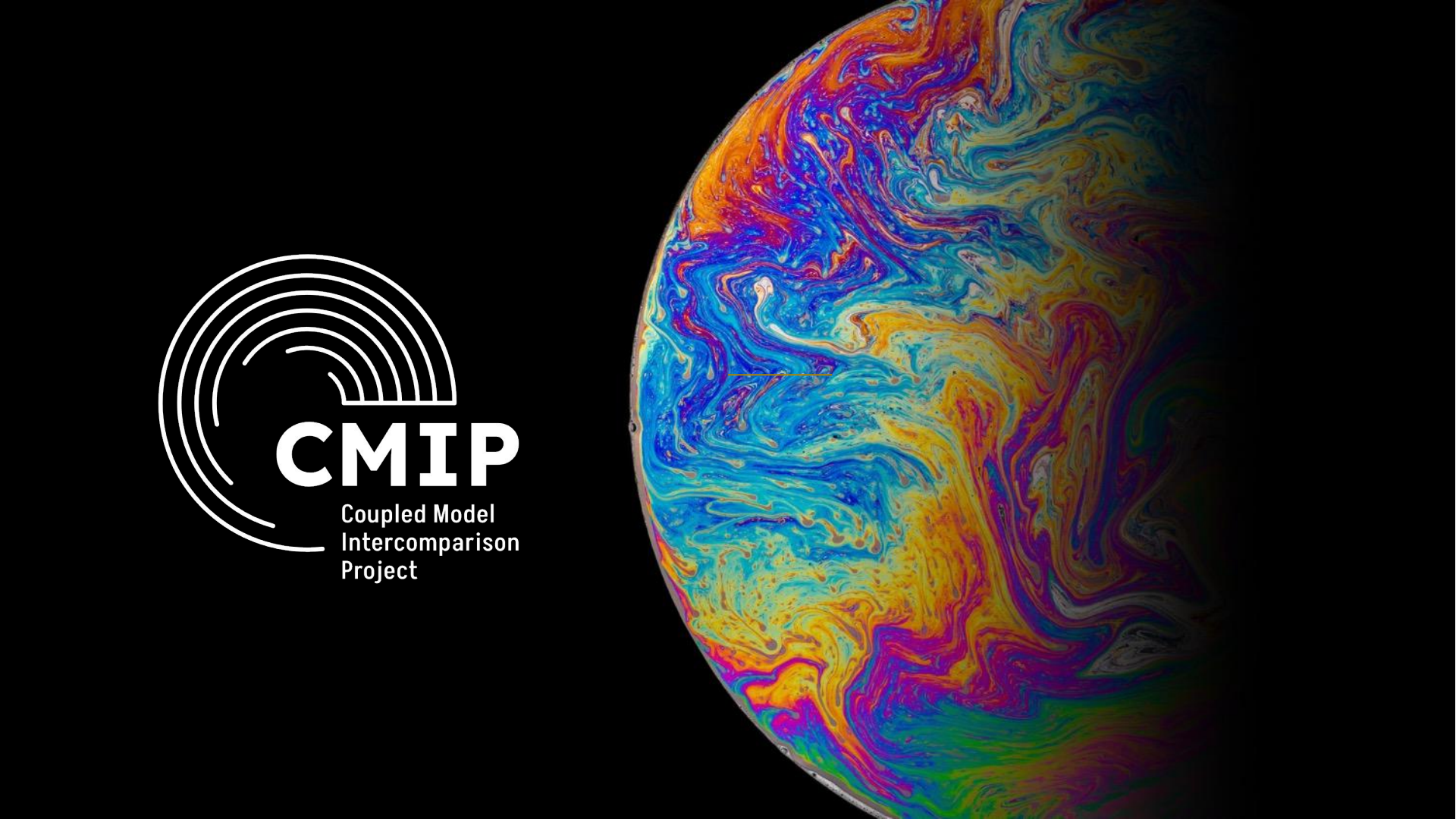


Patterns generated for CMIP6 models, gridded view. Patterns are shown per variable, for the month of...



Distance matrix for temperature, providing the independence metric.

For more examples check https://docs.esmvaltool.org/en/latest/gallery.html


CMIP
Coupled Model
Intercomparison
Project